

Regime-Switching Asset Allocation via a POMDP

Negative headline, decomposed in three diagnostic experiments

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Have you ever wondered...

Why does the “safe” 60/40 portfolio sometimes lose money when you most need it not to?

In **2022**, the textbook 60/40 stock-bond portfolio lost **~17%** as stocks AND bonds fell together.

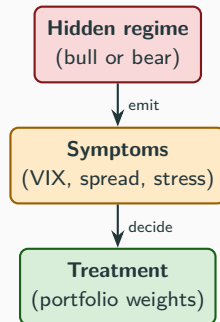
- Static allocation assumes joint return distribution is stable. Reality: conditional heteroskedasticity + stress correlation.
- **Question.** Can a fully decision-theoretic regime overlay beat 60/40 out-of-sample after realistic transaction costs?

The doctor analogy (= POMDP)

You are a doctor. Your patient is the market.

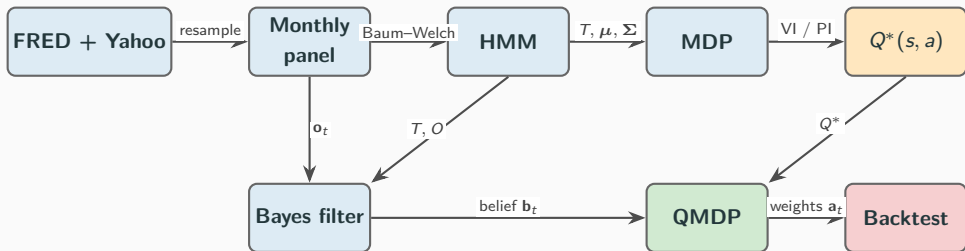
- Patient is *healthy* (bull) or *sick* (bear), but you can't examine directly.
- You only see symptoms (VIX, term spread, stress indices).
- Different treatments (portfolio weights) work better in different states.
- Goal: **read symptoms, infer state, prescribe.**

This IS a POMDP: s_t = regime, $\mathbf{o}_t$ = obs, $\mathbf{a}_t$ = weights, R = expected utility.



*Tree, MDP, ID all fail here.
POMDP is the minimum tool.*

End-to-end pipeline



- **Top (offline, do once):** fit HMM, solve MDP, build Q^* . VI and PI agree to 2.6×10^{-6} .
- **Bottom (each month):** new observation $\rightarrow$ updated belief $\rightarrow$ QMDP $\rightarrow$ weights $\rightarrow$ realised return.
- **Walk-forward** 2003-10 $\rightarrow$ 2026-04, monthly rebalance, 5 bps tx cost, 271 decisions.

QMDP rule, with a worked example

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a}} \sum_s b(s) Q^*(s, \mathbf{a})$$

Decision moment. After a VIX spike the belief filter returns $\mathbf{b} = (0.07, 0.93)$ (bear-leaning). At the *headline* $\gamma=2$, the Q^* table is genuinely regime-differentiated:

Action	$Q^*(\text{bull})$	$Q^*(\text{bear})$	$\mathbf{b}^\top Q^*(\cdot, \mathbf{a})$
100/0 stocks	0.183	0.024	0.035
60/40	0.180	0.028	0.038
0/100 bonds	0.174	0.032	0.042

Bull column rewards stocks ($\text{argmax} = 100/0$); bear column rewards bonds ($\text{argmax} = 0/100$).

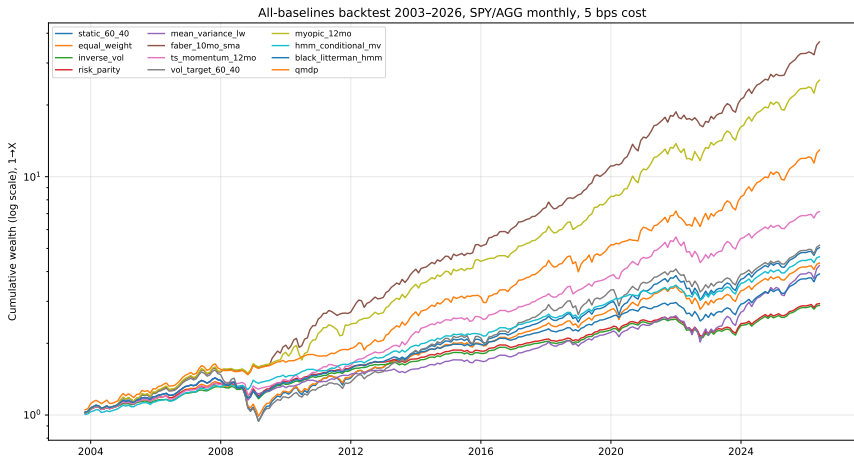
The bear-leaning belief makes QMDP pick **0/100**: a genuine regime-driven decision at the canonical $\gamma=2$.

Twelve-strategy horse race (2003–2026, 5 bps cost)

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Faber 10-mo SMA	17.24%	9.80%	1.68	3.57	−13.5%	1.28
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41
HMM-conditional MV	6.38%	5.91%	1.08	1.78	−18.4%	0.35
Black–Litterman + HMM	6.17%	6.65%	0.94	1.45	−18.0%	0.34
Inverse-vol	4.78%	5.24%	0.92	1.42	−16.8%	0.28
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23
Mean-variance + Ledoit-Wolf	6.58%	8.20%	0.82	1.32	−22.5%	0.29
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
Vol-target 60/40	7.50%	10.14%	0.77	1.12	−39.1%	0.19
QMDP ($\gamma=2$)	10.01%	14.64%	0.73	1.07	−53.0%	0.19

Three sentences. (1) Faber dominates every metric. (2) **QMDP is dead last on Sharpe** with the deepest drawdown. (3) HMM-aware non-QMDP baselines (HMM-MV, BL+HMM) beat QMDP $\Rightarrow$ the HMM signal IS informative.

Equity curves: \$1 invested 2003-10-31 (log scale)



Faber (top) ends near \$30. QMDP (red) ends near \$8 with the deepest mid-figure trench. Static 60/40 ends near \$5.

Why QMDP collapsed: regime label never moves the policy

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80

Within columns: bull and bear give the SAME action at every γ .

Higher γ shifts the policy bondward across-the-board (a vol-reduction effect), but never tilts *between* regimes.

A low- γ “gambler” bets the same way regardless of suspicion. Only $\gamma \geq 8$ would let suspicion shift the policy, and even then, both regimes shift together.

Unlock 1: richer observations FIX the policy at $\gamma=2$

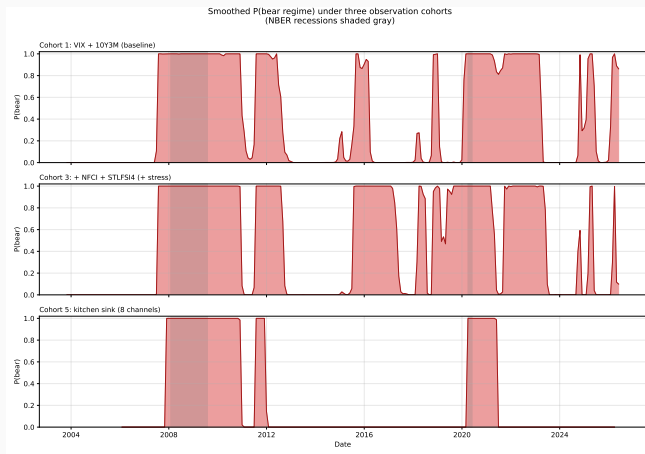
Cohort	Features	Bull	Bear	Differs?
1. Baseline	VIX, T10Y3M	100/0	100/0	$\times$
2. +Yield curve	+T10Y2Y	100/0	100/0	$\times$
3. +Stress	+NFCI, +STLFSI4	100/0	0/100	✓ full flip
4. +Macro	+UMCSENT, +ICSA	100/0	0/100	✓
5. Kitchen sink	8 channels	100/0	60/40	✓

Same $\gamma=2$ as the headline. We just changed which observations feed the HMM.

Adding **NFCI** (Chicago Fed) + **STLFSI4** (St. Louis Fed) = **full policy flip**.

“More thermometers, better diagnosis.” The collapse is an observation-channel problem.

Visual confirmation of Unlock 1



Top: cohort 1 (baseline, noisy false alarms). Middle: cohort 3 (+NFI, +STLFSI4), with sharp bear spikes clearly bracketing every NBER recession. Bottom: cohort 5 (kitchen sink, over-fits).

Unlock 2: walk-forward refit lifts Sharpe to 1.08

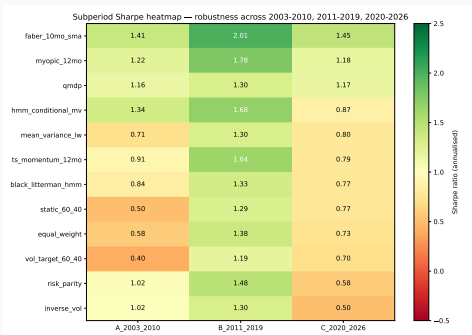
Variant	CAGR	Sharpe	Max DD	Calmar
Static 60/40 (benchmark)	7.40%	0.81	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	0.96	−34.2%	0.29
QMDP annual refit (5y win)	9.11%	0.85	−25.1%	0.36
QMDP quarterly refit	8.56%	0.83	−23.4%	0.37
QMDP expanding window	9.81%	1.08	−16.5%	0.60
HMM-MV expanding window	5.32%	0.95	−16.8%	0.32

Following Nystrup–Madsen–Lindström (2018):

Sharpe 0.81 → **1.08** (+33%) · **Max DD** −34% → **−16.5%** (more than halves)
· **Calmar** 0.22 → **0.60** (nearly triples)

“Re-read the casino’s recent hands.” A fixed HMM was bringing stale priors.

Subperiod robustness: QMDP fails worst where it should help



- **Faber dominates every column.** The trend rule isn't a one-period fluke.
- **QMDP's worst column is Period A (GFC, Sharpe 0.33):** exactly where a regime model should help most.
- In the same Period A, HMM-MV and BL+HMM (same signal, no QMDP projection) get Sharpe 1.03–1.25 $\Rightarrow$ **QMDP's $\gamma=2$ projection destroyed it.**

The honest verdict: claim vs. non-claim

We DO claim	We do NOT claim
POMDP is the right <i>minimum</i> tool for this decision problem.	POMDP asset allocation beats simple trend-following.
The headline negative finding is a diagnostic of methodology, not fundamentals.	The original QMDP at $\gamma=2$ is a useful production policy.
The HMM signal is informative (HMM-MV, BL+HMM beat QMDP).	Our walk-forward QMDP is the new state-of-the-art.
Three components separately reverse the negative finding: observations, refit, γ .	Our gains net of transaction costs matter to a real allocator.

Negative-with-decomposition is more useful than a positive headline. The diagnostic table outlasts any specific sample's Sharpe number.

Three extensions in priority order

1. **CVaR-penalised reward.**

Replace CRRA utility with $R = \mathbb{E}[r] - \kappa \text{CVaR}_\alpha[r]$ to explicitly penalise bear-regime tail risk. One-line change in the reward function.

2. **Point-based value iteration (PBVI).**

Maintain explicit α -vectors over a small set of reachable beliefs. Gives a tighter upper bound on QMDP. On a 1-D belief simplex the cost is negligible.

3. **Off-policy MC + Double Q-learning.**

A genuinely model-free baseline that does *not* assume the QMDP transition matrix is correct. Cross-checks the model-based pipeline against learning from scratch.

Thank you.

Questions?